FREE-ENERGY FLUCTUATIONS OF MIXED p-SPIN MODELS THROUGHOUT THE HIGH-TEMPERATURE REGIME

ABSTRACT. We prove Gaussian fluctuations of the free energy for finite, zero-field, distinct-index mixed p-spin models whose minimum active interaction order is at least three. The result holds at every fixed inverse temperature strictly below the thermodynamic critical point. Uniformly on compact subintervals of this regime, the free energy admits a quadratic approximation in the Gaussian couplings with an explicit vanishing mean-square error at the fluctuation scale. The proof combines Chen's high-temperature overlap bounds with Gaussian interpolation and the Poincaré inequality. The limiting variance is determined by the smallest active interaction order, and the centering retains the exact finite-size covariance.

CONTENTS

1. Introduction and main results 1
2. Proof of Theorem 1.1 6
3. Proof of the central limit theorem 13
References 14

1. INTRODUCTION AND MAIN RESULTS

The limiting free energy describes the typical thermodynamic behavior of a spin glass, but does not determine the size or origin of its sample-to-sample fluctuations. For interactions involving at least three distinct spins, these fluctuations can be much smaller than the finite-size corrections to the limiting free energy. A central question is whether their leading contribution can nevertheless be described by a simple statistic of the random couplings.

This paper studies that question for finite mixtures without an external field. We identify the leading quadratic contribution throughout the strict thermodynamic high-temperature regime and obtain a Gaussian limit with the exact finite-size centering. The approximation also separates two roles of the mixture: the smallest active interaction order determines the leading random scale, whereas the whole covariance function determines the high-temperature region.

1.1. **Setting.** We first introduce the definition of the mixed p-spin models. Fix an integer $P \geqslant 3$ and real numbers $\gamma_3, \dots, \gamma_P \in \mathbb{R}$ such that $\sum_{i=3}^{P} |\gamma_i| \neq 0$. Let

$$\mathcal{P} = \{p \in \{3, \dots, P\} : \gamma_p \neq 0\}, \qquad p_\star = \min \mathcal{P}. \tag{1}$$

2020 *Mathematics Subject Classification.* Primary: 82B26, 82B44, 60F05.
Key words and phrases. Spin glasses, high temperature, free-energy fluctuations, central limit theorem.

For $N \geqslant P$, let $\Sigma_N = \{-1,+1\}^N$ be the configuration space. For each $\sigma \in \Sigma_N$, the mixed p-spin Hamiltonian is

$$H_N(\sigma) = \sum_{p\in\mathcal{P}} \frac{\gamma_p}{N^{(p-1)/2}} \sum_{I\in\binom{[N]}{p}} g_{p,I}\sigma_I, \tag{2}$$

where for every $p \in \mathcal{P}$ and every p-hyperedge $I \in \binom{[N]}{p}$, $g_{p,I}$ are independent standard Gaussian random variables, and $\sigma_I = \prod_{i\in I}\sigma_i$. For two configurations $\sigma^1, \sigma^2 \in \Sigma_N$, define their overlap by

$$R_{1,2} = \frac{1}{N}\sum_{i=1}^{N} \sigma_i^1\sigma_i^2. \tag{3}$$

For each $p \in \mathcal{P}$, define the finite-N distinct-index covariance kernel by

$$\kappa_{p,N}(R_{1,2}) = \frac{1}{N^p} \sum_{I\in\binom{[N]}{p}} \sigma_I^1\sigma_I^2. \tag{4}$$

The quantity on the right-hand side depends on (σ^1, σ^2) only through $R_{1,2}$. A simple calculation gives

$$\mathbb{E}\, H_N(\sigma^1)H_N(\sigma^2) = N\xi_N(R_{1,2}). \tag{5}$$

where

$$\xi_N(q) := \sum_{p\in\mathcal{P}} \gamma_p^2\kappa_{p,N}(q). \tag{6}$$

In particular,

$$\xi_N(1) = \sum_{p\in\mathcal{P}} \gamma_p^2 \frac{\binom{N}{p}}{N^p}. \tag{7}$$

The limiting covariance function is

$$\xi(q) = \sum_{p\in\mathcal{P}} \frac{\gamma_p^2}{p!}q^p, \qquad q \in [-1,1]. \tag{8}$$

In particular,

$$\xi(1) = \sum_{p\in\mathcal{P}} \frac{\gamma_p^2}{p!}. \tag{9}$$

For $\beta \geqslant 0$, define the normalized partition function and the free energy respectively

$$Z_N(\beta) = 2^{-N} \sum_{\sigma\in\Sigma_N} \exp\left\{\beta H_N(\sigma)\right\}, \qquad F_N(\beta) = \frac{1}{N}\log Z_N(\beta). \tag{10}$$

Since

$$\mathbb{E}\, H_N(\sigma)^2 = N\xi_N(1)$$

for every $\sigma \in \Sigma_N$, one has

$$\frac{1}{N}\log \mathbb{E}\, Z_N(\beta) = \frac{\beta^2}{2}\xi_N(1). \tag{11}$$

Let

$$F(\beta) = \lim_{N\to\infty} \mathbb{E}\, F_N(\beta) \tag{12}$$

denote the limiting free energy. We define the critical inverse temperature by

$$\beta_c(\xi) = \sup\left\{b \geqslant 0 : F(\beta) = \frac{\beta^2}{2}\xi(1) \text{ for every } 0 \leqslant \beta \leqslant b\right\}. \tag{13}$$

Thus the strict high-temperature regime considered in this paper is

$$0 \leqslant \beta < \beta_c(\xi). \tag{14}$$

All estimates below are uniform on compact intervals

$$0 \leqslant \beta \leqslant b, \qquad b < \beta_c(\xi). \tag{15}$$

We do not include the endpoint $\beta = \beta_c(\xi)$.

Define the quadratic disorder statistic

$$J_N(\beta) = \frac{\beta^2}{2} \sum_{p \in \mathcal{P}} \frac{\gamma_p^2}{N^p} \sum_{I \in \binom{[N]}{p}} g_{p,I}^2. \tag{16}$$

Its expectation is

$$\mathbb{E}\, J_N(\beta) = \frac{\beta^2}{2} \xi_N(1). \tag{17}$$

1.2. **Main results.** Our first result identifies the quadratic disorder statistic $J_N(\beta)$ as the complete leading random contribution to the free energy throughout the strict high-temperature regime.

Theorem 1.1 (Quadratic approximation)**.** *Fix $0 \leqslant b < \beta_c(\xi)$. There exists a constant $C = C(b, \xi) < \infty$ such that, for every $N \geqslant P$,*

$$\sup_{0 \leqslant \beta \leqslant b} \mathbb{E}\left[\left| N^{p_\star/2} \left(F_N(\beta) - J_N(\beta) \right) \right|^2 \right] \leqslant C N^{1 - p_\star/2}. \tag{18}$$

Consequently,

$$\lim_{N \to \infty} \sup_{0 \leqslant \beta \leqslant b} \mathbb{E}\left[\left| N^{p_\star/2} \left(F_N(\beta) - J_N(\beta) \right) \right|^2 \right] = 0. \tag{19}$$

The second result gives the leading free-energy fluctuation. The fluctuation scale and limiting variance are determined by the smallest active interaction degree $p_\star$.

Theorem 1.2 (CLT of free energy)**.** *For every fixed*

$$0 < \beta < \beta_c(\xi),$$

one has

$$N^{p_\star/2} \left(F_N(\beta) - \frac{\beta^2}{2} \xi_N(1) \right) \Longrightarrow \mathcal{N}\left(0, \frac{\beta^4 \gamma_{p_\star}^4}{2 p_\star!} \right). \tag{20}$$

The finite-N centering in Theorem 1.2 is essential. Indeed, expanding the falling factorial gives

$$\xi_N(1) - \xi(1) = -\frac{1}{N} \sum_{p \in \mathcal{P}} \frac{\gamma_p^2}{p!} \binom{p}{2} + O(N^{-2}), \tag{21}$$

whereas the random fluctuation scale in (20) is $N^{-p_\star/2}$. Since $p_\star \geqslant 3$, the coefficient of N^{-1} in (21) is strictly negative for a nontrivial mixture. Thus the deterministic centering error has order N^{-1}, which is larger than the fluctuation scale for each fixed $\beta > 0$.

1.3. **Related work and motivation.** The thermodynamic limit and the fluctuations around it are different questions. Talagrand's proof of the Parisi formula [18] and Panchenko's extension to mixed models with odd interactions [16] provide the variational description of the limiting free energy. They do not, by themselves, specify a central limit scale or the finite-size centering at that scale. The definition of $\beta_c(\xi)$ above uses this thermodynamic limit. A main issue in fluctuation theory is whether estimates obtained in an explicit sufficient high-temperature range continue to hold throughout the region where the quenched and annealed limits agree.

For the zero-field Sherrington–Kirkpatrick model, Aizenman, Lebowitz, and Ruelle [1] established high-temperature Gaussian fluctuations through a graphical expansion whose leading terms are loops. Comets and Neveu [9] developed a different approach using Brownian motions and continuous martingales. These two methods lead to distinct ways of studying higher-order interactions: one identifies the combinatorial structures contributing to the partition function, while the other controls a stochastic representation of its fluctuations. Both approaches require estimates that are sharper than concentration of the free energy around its thermodynamic limit.

For pure distinct-index models with $p \geqslant 3$, Bovier, Kurkova, and Löwe [4] combined martingale arguments with truncation to obtain a Gaussian limit for $\log(Z_N/\mathbb{E}\,Z_N)$ on the scale $N^{-(p-2)/2}$ in an explicit high-temperature range. The corresponding scale for the free energy per spin is $N^{-p/2}$. Knöpfel and Löwe [15] extended the pure-model conclusion to symmetric, variance-one couplings with an exponential moment, again in a specified high-temperature range. Their factorization into a product of hyperbolic cosines and a remaining spin average shows why the fluctuations of squared couplings enter. Thus both the pure-model Gaussian limit and its leading quadratic mechanism have direct precedents.

The temperature restriction in these results deserves separate attention. Talagrand [17] used a truncated second-moment argument to obtain an explicit lower bound on the critical inverse temperature of the pure model. Bovier and Schertzer [5] subsequently established the pure-model fluctuation theorem throughout the range given by that bound. They also identify the quadratic coupling statistic as the complete leading contribution and prove a further central limit theorem after subtracting it. Their paper explicitly distinguishes this threshold from the actual critical inverse temperature. It uses a normalization in which the Hamiltonian variance is exactly N. The comparison with the present theorem therefore involves the temperature range, the treatment of mixtures, and the strength of the approximation, as well as the finite-size convention.

Mixed models with a two-spin component have a different leading structure. Banerjee and Belius [2] prove a Gaussian limit at sufficiently high temperature for canonical mixed models with covariance $N\xi(R_{1,2})$ and a nonzero two-spin coefficient. The order-one fluctuations of $\log Z_N$ arise from weighted cycles formed from that component. Consequently, the assumption $p_\star \geqslant 3$ in the present paper separates two different fluctuation mechanisms; it is not a restriction that can be removed by inserting a two-spin term into the variance formula.

The closest graphical treatment of higher-order mixtures is due to Dey and Wu [11]. They study distinct-index mixed models with symmetric disorder under moment assumptions and with weak external fields. Their decomposition separates the product of hyperbolic cosines from an interacting factor, whose fluctuations are described by leading hypergraphs. At zero field, the relevant structures depend on the smallest active even and odd orders. Their

fluctuation analysis works below the explicit second-moment threshold

$$\beta_f^2 = \inf_{0<q<1} \frac{I(q)}{\xi(q)}, \qquad I(q) = \frac{(1+q)\log(1+q) + (1-q)\log(1-q)}{2}.$$

This threshold is not identified with $\beta_c(\xi)$. Their results resolve the interacting contribution and allow disorder and fields beyond the Gaussian zero-field setting considered here.

A second line of work studies fluctuations through overlap estimates. For the pure p-spin model with a field, Bardina, Márquez-Carreras, Rovira, and Tindel [3] developed cavity estimates that bound centered overlap moments at their natural orders when the inverse temperature is sufficiently small. Such estimates give quantitative control of replica-symmetric behavior, but their temperature hypothesis is stronger than merely lying below the thermodynamic critical point. For the zero-field higher-order models relevant here, Chen [8] proves overlap moment bounds throughout the strict thermodynamic high-temperature region; Theorem 4 and the mixed-model extension in Remark 3 are the specific inputs used below.

Chen's theorem is stated for the canonical covariance, so its application to (2) requires a comparison. After matching the normalization, we transfer its bounds to the distinct-index model by replicated Gaussian interpolation. The remaining argument combines cancellations in the finite-index covariance kernels with Gaussian integration by parts and the Poincaré inequality. This gives a mean-square estimate for the error after removing the quadratic statistic. The central limit theorem then follows from an independent sum, rather than from a classification of the leading interacting hypergraphs.

External fields illustrate why one must specify the regime before comparing fluctuation theorems. Guerra and Toninelli [14] obtained Gaussian fluctuations for the SK model with a nonzero field in a sufficiently high-temperature region using quadratic replica coupling and cavity equations. Tindel [19] gave a stochastic-calculus treatment in a sufficiently high-temperature region. For mixed models, Chen, Dey, and Panchenko [7] prove variance of order N for the extensive free energy when the field is nonzero; for mixtures without odd interaction terms of order at least three, they also prove a central limit theorem at every temperature. The resulting $N^{-1/2}$ scale for the free energy per spin differs from the zero-field scale in Theorem 1.2.

The passage from zero field to a fixed nonzero field is itself a fluctuation problem. Dey and Wu [13] study SK models with $h_N = \rho N^{-\alpha}$ and identify a transition at $\alpha = 1/4$: weaker fields leave the leading zero-field law unchanged, the boundary scale changes its parameters, and stronger vanishing fields produce a larger fluctuation scale. For $\alpha \geqslant 1/4$, their results cover every fixed $\beta < 1$ and symmetric disorder with a finite sixth moment. For $\alpha < 1/4$, the stated theorem assumes Gaussian disorder and sufficiently small β. Their expansion introduces paths in addition to loops. These results explain why a theorem at zero field does not automatically describe a sequence of small nonzero fields.

Inhomogeneity across species supplies another useful comparison. Dey and Wu [12] develop a species-wise cavity method for the multispecies SK model, including indefinite interaction matrices. They establish overlap concentration and a positive-field free-energy central limit theorem in a specified high-temperature region; their zero-field overlap bound extends to the spectral threshold. Their weak-field work [13] also treats multispecies fluctuations, with the bipartite case developed in detail. Species-dependent two-spin variances lead to vector overlaps. They differ from a mixture of interaction orders acting on the same population of spins, for which the scalar covariance ξ used here is the relevant object.

Two further distinctions concern the strength and the uniformity of fluctuation estimates. Chatterjee's superconcentration theorem [6] gives a sublinear variance bound for the extensive

zero-field SK free energy at every temperature; such a bound alone does not identify its precise limiting law. At the high-temperature boundary, Dey and Kang [10] prove a variance asymptotic and a Gaussian limit for SK inverse temperatures approaching 1 with $N^{1/3}(1-\beta_N^2)$ bounded away from zero. This requires estimates sensitive to the distance from criticality. The compact-interval uniformity in Theorem 1.1 does not provide such an N-dependent approach to $\beta_c(\xi)$.

Finally, thermodynamic equivalence is insufficient for transferring these small fluctuation scales. Bovier, Kurkova, and Löwe [4] also analyze a repeated-index convention and obtain a different scale. The exact-variance normalization of [5] likewise differs from (2) by an N-dependent factor. Here the annealed center is exactly $\beta^2\xi_N(1)/2$; replacing it by $\beta^2\xi(1)/2$ introduces the nonzero correction of order N^{-1} in (21), which is larger than the random scale. Retaining that correction is necessary for the stated limit.

The present result places the established quadratic mechanism in a finite-mixture theorem with exact centering and a uniform mean-square error bound throughout every fixed compact subinterval of the strict thermodynamic high-temperature region. Only the smallest active interaction order survives in the limiting variance, whereas the whole mixture can affect $\beta_c(\xi)$. The uniformity is the supremum, over temperature, of the mean-square error in (18). No convergence of a temperature-indexed process, critical-endpoint limit, or second-scale theorem after subtraction of J_N is asserted.

2. Proof of Theorem 1.1

The case $b = 0$ is immediate. Fix $0 < b < \beta_c(\xi)$. All constants in this section are chosen on the entire interval $0 \leqslant s \leqslant b^2$ and may depend on b and the fixed mixture, but not on N or on the evaluation point $\beta_0 \in [0, b]$. For $s \in [0, b^2]$, let $\langle\cdot\rangle_s$ denote the Gibbs expectation associated with the Hamiltonian $\sqrt{s}\, H_N(\sigma)$, namely

$$G_{N,s}(\sigma) = \frac{2^{-N}\exp\{\sqrt{s}\, H_N(\sigma)\}}{Z_N(\sqrt{s})}. \tag{22}$$

For a function $f : \Sigma_N^{\otimes\ell} \to \mathbb{R}$ with ℓ finite, write

$$\nu_s(f) = \mathbb{E}\langle f\rangle_s. \tag{23}$$

At $s = 0$, the associated Gibbs measure is uniform on Σ_N. The corresponding expectation will also be denoted by ν_0. For an arbitrary $\beta_0 \in [0, b]$, set

$$D_N = \log Z_N(\beta_0) - N J_N(\beta_0). \tag{24}$$

The scaled error in Theorem 1.1 is given by the identity

$$N^{p_\star/2}\big(F_N(\beta_0) - J_N(\beta_0)\big) = N^{p_\star/2-1} D_N. \tag{25}$$

We will control both the mean and the variance of D_N through the same quantity. Define

$$A_N(s) = \nu_s\big(\xi_N(R_{1,2})\big), \qquad 0 \leqslant s \leqslant b^2, \tag{26}$$

and

$$V_N = \nu_0\big(\xi_N(R_{1,2})^2\big) = \sum_{p\in\mathcal{P}} \gamma_p^4 \frac{\binom{N}{p}}{N^{2p}}. \tag{27}$$

The second identity in (27) follows from the orthogonality of distinct Walsh characters under the uniform measure. In particular,

$$V_N = O\big(N^{-p_\star}\big). \tag{28}$$

Also, using two independent Gibbs replicas,

$$A_N(s) = \sum_{p\in\mathcal{P}} \frac{\gamma_p^2}{N^p} \sum_{I\in\binom{[N]}{p}} \mathbb{E}\langle\sigma_I\rangle_s^2 \geqslant 0. \tag{29}$$

The central estimate of the section is the following linear-response statement.

Proposition 2.1 (Linear response of the covariance kernel)**.** *For the fixed bound $b < \beta_c(\xi)$, one has*

$$A_N(s) = sNV_N + O_{b,\xi}\big(N^{2-3p_\star/2}\big) \tag{30}$$

uniformly for $0 \leqslant s \leqslant b^2$.

The need for the precise leading term in (30) comes from the Gaussian Poincaré estimate used at the end of the section. Its two principal terms are

$$\beta_0^2 N A_N(\beta_0^2) - \beta_0^4 N^2 V_N. \tag{31}$$

Proposition 2.1 makes the leading parts of the two terms in (31) cancel exactly. The next three subsections are devoted to the proof of Proposition 2.1. We return to D_N and prove Theorem 1.1 only after that proposition has been established.

2.1. **Canonical comparison and overlap moments.** The proof of Proposition 2.1 requires a microscopic moment bound for the overlap under the distinct-index Gibbs measure. The available full-high-temperature estimate is formulated for the canonical mixed p-spin model with covariance $N\xi(R_{1,2})$. We therefore introduce that model only as an auxiliary device for importing the overlap estimate. We do not use it to compare the microscopic free-energy fluctuations of the two models.

Let $\widetilde{H}_N$ be a centered Gaussian process on Σ_N, independent of H_N, with covariance

$$\mathbb{E}\,\widetilde{H}_N(\sigma^1)\widetilde{H}_N(\sigma^2) = N\xi(R_{1,2}). \tag{32}$$

The following elementary comparison is the reason the canonical model can be used.

Lemma 2.2 (Bounded covariance difference)**.** *There exists $C_\xi < \infty$ such that*

$$\sup_{\sigma^1,\sigma^2\in\Sigma_N} |N\xi_N(R_{1,2}) - N\xi(R_{1,2})| \leqslant C_\xi \tag{33}$$

for every $N \geqslant P$.

Proof. Fix $p \in \mathcal{P}$ and put

$$x_i = \sigma_i^1\sigma_i^2, \qquad r = \frac{1}{N}\sum_{i=1}^N x_i.$$

Let $I_1, \dots, I_p$ be independent and uniform on $[N]$, and let $\mathcal{D}$ be the event that these indices are all distinct. Then

$$\mathbb{E}\prod_{a=1}^p x_{I_a} = r^p,$$

whereas

$$\mathbb{E}\left[\prod_{a=1}^p x_{I_a}\,\mathbf{1}_{\mathcal{D}}\right] = \frac{p!}{N^p}\sum_{I\in\binom{[N]}{p}}\prod_{i\in I} x_i = p!\,\kappa_{p,N}(r).$$

Since the product has absolute value one,

$$\left|\kappa_{p,N}(r) - \frac{r^p}{p!}\right| \leqslant \frac{1}{p!}\,\mathbb{P}(\mathcal{D}^c) \leqslant \frac{1}{p!}\frac{\binom{p}{2}}{N}. \tag{34}$$

Multiplying by $N\gamma_p^2$, summing over the finite set $\mathcal{P}$, and using the definitions of ξ_N and ξ proves (33). ∎

For later use, set

$$\Delta_N(\sigma,\tau) = N\xi_N(R(\sigma,\tau)) - N\xi(R(\sigma,\tau)). \tag{35}$$

Then Lemma 2.2 states that $\|\Delta_N\|_\infty \leqslant C_\xi$.

For completeness, Lemma 2.2 also shows that the distinct-index and canonical models have the same limiting free energy. Let

$$\Phi_N(t) = \frac{1}{N}\,\mathbb{E}\log\left(2^{-N}\sum_{\sigma\in\Sigma_N}\exp\left\{\beta\left(\sqrt{t}\,H_N(\sigma) + \sqrt{1-t}\,\widetilde{H}_N(\sigma)\right)\right\}\right).$$

Gaussian integration by parts gives

$$\Phi_N'(t) = \frac{\beta^2}{2N}\,\mathbb{E}\left\langle \Delta_N(\sigma^1,\sigma^1) - \Delta_N(\sigma^1,\sigma^2)\right\rangle_t.$$

Consequently, for every fixed β,

$$\left|\mathbb{E}\,F_N(\beta) - \mathbb{E}\,\widetilde{F}_N(\beta)\right| \leqslant \frac{\beta^2 C_\xi}{N}. \tag{36}$$

Thus the critical inverse temperature defined in terms of the limiting free energy is the same for both models.

We next record the overlap input from Theorem 4 and Remark 3 of [8]. To match conventions, put $a = \xi(1) > 0$ and $c_p = \gamma_p^2/(p!a)$. Then $c_p \geqslant 0$, $\sum_p c_p = 1$, and Chen's normalized covariance is $\xi_C(q) = \frac{1}{2}\sum_p c_p q^p = \xi(q)/(2a)$. The canonical model with covariance $N\xi$ at inverse temperature $\sqrt{s}$ therefore corresponds to the normalized model at inverse temperature $\sqrt{2as}$. The annealed free energies and critical inverse temperatures scale by the same change of variable. The thermodynamic limit for this canonical model is the background limit used in that result; the comparison above transfers it to the distinct-index model. The cited theorem and remark imply that for every integer $k \geqslant 1$ there exists $C = C(b,k,\xi)$ such that

$$\sup_{0\leqslant s\leqslant b^2}\widetilde{\nu}_s\big(|R_{1,2}|^{2k}\big) \leqslant \frac{C}{N^k}. \tag{37}$$

Here $\widetilde{\nu}_s$ denotes the disorder-averaged Gibbs expectation for the canonical Hamiltonian $\sqrt{s}\,\widetilde{H}_N$. The assumption $p_\star \geqslant 3$ is exactly the lowest-degree hypothesis in Remark 3 of [8].

We now transfer (37) to the model of this paper.

Lemma 2.3 (Overlap moments for the distinct-index model)**.** *For every integer $k \geqslant 1$, there exists $C = C(b,k,\xi) < \infty$ such that*

$$\sup_{0\leqslant s\leqslant b^2}\nu_s\big(|R_{1,2}|^{2k}\big) \leqslant \frac{C}{N^k}. \tag{38}$$

Proof. For $t \in [0,1]$, define the interpolating Hamiltonian

$$H_{N,t}(\sigma) = \sqrt{t}\, H_N(\sigma) + \sqrt{1-t}\, \widetilde{H}_N(\sigma), \tag{39}$$

and let $\nu_{s,t}$ be the disorder-averaged replicated Gibbs expectation associated with $\sqrt{s}\, H_{N,t}$. Recall from (35) that

$$\|\Delta_N\|_\infty \leqslant C_\xi.$$

We use the replicated Gaussian interpolation identity. If f is a bounded observable of n replicas, then

$$\frac{d}{dt}\nu_{s,t}(f) = \frac{s}{2}\nu_{s,t}\Bigg(f\Bigg[\sum_{\ell,\ell'=1}^{n}\Delta_{\ell,\ell'} - 2n\sum_{\ell=1}^{n}\Delta_{\ell,n+1} + n(n+1)\Delta_{n+1,n+2} - n\Delta_{n+1,n+1}\Bigg]\Bigg), \tag{40}$$

where

$$\Delta_{\ell,\ell'} = \Delta_N(\sigma^\ell, \sigma^{\ell'}).$$

The last diagonal term in (40) is part of the general identity. In the present model all diagonal values of Δ_N are equal, so the diagonal terms can also be canceled and (40) can be written in the usual pair-overlap form. Formula (40) follows by differentiating the replicated Gibbs expectation and applying Gaussian integration by parts to each of the two Gaussian processes in (39).

Apply (40) with $n = 2$ and $f = |R_{1,2}|^{2k}$. Since $f \geqslant 0$, $s \leqslant b^2$, and $\|\Delta_N\|_\infty \leqslant C_\xi$, there exists $C = C(b,k,\xi)$ such that

$$\left|\frac{d}{dt}\nu_{s,t}(f)\right| \leqslant C\nu_{s,t}(f). \tag{41}$$

Gronwall's inequality, followed by (37), gives

$$\nu_{s,1}(f) \leqslant e^C \nu_{s,0}(f) \leqslant \frac{C'}{N^k}$$

uniformly for $0 \leqslant s \leqslant b^2$. Since $\nu_{s,1} = \nu_s$, this proves (38). ∎

2.2. **From overlap moments to covariance-kernel moments.** The next lemma gives a finite-N bound on the distinct-index kernel.

Lemma 2.4 (Finite-N polynomial bound)**.** *For every fixed integer $p \geqslant 1$, there exists $C_p < \infty$ such that, for every $x_1, \dots, x_N \in \{-1, +1\}$ and $r = N^{-1}\sum_{i=1}^N x_i$,*

$$\left|\frac{1}{N^p}\sum_{I \in \binom{[N]}{p}}\prod_{i\in I} x_i\right| \leqslant C_p \sum_{j=0}^{\lfloor p/2\rfloor} N^{-j}|r|^{p-2j}. \tag{42}$$

Proof. Let $e_p(x_1, \dots, x_N)$ be the pth elementary symmetric polynomial. For a partition $\lambda \vdash p$, write $\ell(\lambda)$ for its number of parts, $o(\lambda)$ for the number of odd parts, and

$$z_\lambda = \prod_{m \geqslant 1} m^{a_m(\lambda)} a_m(\lambda)!,$$

where $a_m(\lambda)$ is the multiplicity of the part m. The standard power-sum expansion of the elementary symmetric polynomial is

$$e_p = \sum_{\lambda \vdash p} (-1)^{p-\ell(\lambda)} z_\lambda^{-1} \prod_{a=1}^{\ell(\lambda)} \left(\sum_{i=1}^{N} x_i^{\lambda_a} \right). \tag{43}$$

Since $x_i \in \{-1, +1\}$,

$$\sum_{i=1}^{N} x_i^m = \begin{cases} Nr, & m \text{ odd}, \\ N, & m \text{ even}. \end{cases}$$

Consequently, the absolute value of the term indexed by λ in $N^{-p} e_p$ is

$$z_\lambda^{-1} N^{\ell(\lambda)-p} |r|^{o(\lambda)}. \tag{44}$$

Set $j = (p - o(\lambda))/2$. The parity of a partition gives $j \in \{0, \dots, \lfloor p/2 \rfloor\}$ and $o(\lambda) = p - 2j$. Moreover,

$$p - \ell(\lambda) = \sum_{a=1}^{\ell(\lambda)} (\lambda_a - 1) \geqslant \frac{p - o(\lambda)}{2} = j. \tag{45}$$

Indeed, an odd part m contributes $(m-1)/2$ to $(p-o(\lambda))/2$ and an even part m contributes $m/2$, while both are at most $m-1$. Thus (44) is bounded by $z_\lambda^{-1} N^{-j} |r|^{p-2j}$. Summing over the finitely many partitions of p proves (42). ∎

We now obtain the only kernel moment estimate needed below.

Lemma 2.5 (Cubic covariance-kernel moment)**.** *There exists $C = C(b, \xi) < \infty$ such that*

$$\sup_{0 \leqslant s \leqslant b^2} \nu_s \left(|\xi_N(R_{1,2})|^3 \right) \leqslant C N^{-3p_\star/2}. \tag{46}$$

Proof. Fix $p \in \mathcal{P}$. By Lemma 2.4,

$$|\kappa_{p,N}(R_{1,2})| \leqslant C_p \sum_{j=0}^{\lfloor p/2 \rfloor} N^{-j} |R_{1,2}|^{p-2j}.$$

Choose an integer k so large that $2k \geqslant 3P$. By Lemma 2.3,

$$\sup_{0 \leqslant s \leqslant b^2} \|R_{1,2}\|_{L^{2k}(\nu_s)} \leqslant C N^{-1/2}.$$

Since $3(p-2j) \leqslant 2k$, monotonicity of L^q norms gives

$$\sup_{0 \leqslant s \leqslant b^2} \left\| N^{-j} R_{1,2}^{p-2j} \right\|_{L^3(\nu_s)} \leqslant C N^{-j} N^{-(p-2j)/2} = C N^{-p/2}. \tag{47}$$

The case $p - 2j = 0$ is included in the same bound. Hence

$$\sup_{0 \leqslant s \leqslant b^2} \|\kappa_{p,N}(R_{1,2})\|_{L^3(\nu_s)} \leqslant C N^{-p/2}.$$

Using the finite-mixture representation $\xi_N = \sum_{p \in \mathcal{P}} \gamma_p^2 \kappa_{p,N}$ and the triangle inequality in L^3, we obtain

$$\sup_{0 \leqslant s \leqslant b^2} \|\xi_N(R_{1,2})\|_{L^3(\nu_s)} \leqslant C N^{-p_\star/2},$$

which is equivalent to (46). ∎

2.3. **Proof of the linear-response proposition.** We first record the stochastic-stability identity used in the proof.

Lemma 2.6 (Stochastic stability)**.** *Let f be a bounded observable of n replicas that does not depend on the disorder. With*

$$\xi_{\ell,\ell'} = \xi_N(R_{\ell,\ell'}),$$

one has, for $s > 0$,

$$\frac{d}{ds}\nu_s(f) = N\nu_s\left(f\left[\sum_{1\leqslant\ell<\ell'\leqslant n}\xi_{\ell,\ell'} - n\sum_{\ell=1}^{n}\xi_{\ell,n+1} + \frac{n(n+1)}{2}\xi_{n+1,n+2}\right]\right). \tag{48}$$

The identity extends to $s = 0$ by continuity.

Proof. Differentiate the replicated Gibbs expectation associated with $\sqrt{s}\,H_N$ and apply Gaussian integration by parts. In the general formula, the diagonal covariance terms are

$$\frac{1}{2}\left(\sum_{\ell=1}^{n} N\xi_N(1) - nN\xi_N(1)\right) = 0,$$

because the diagonal covariance is constant. The remaining pair-overlap terms are exactly those in (48). For each fixed N, the spin space is finite and the differentiated Gibbs observables have integrable Gaussian bounds. After integration by parts, the displayed right-hand side is continuous at $s = 0$, which justifies the stated endpoint extension. ∎

Proof of Proposition 2.1. Since $A_N(s) = \nu_s(\xi_{1,2})$, applying Lemma 2.6 with $n = 2$ gives

$$A_N'(s) = N\nu_s\left(\xi_{1,2}^2 - 4\xi_{1,2}\xi_{1,3} + 3\xi_{1,2}\xi_{3,4}\right). \tag{49}$$

Introduce

$$B_{1,N}(s) = \nu_s(\xi_{1,2}^2), \tag{50}$$

$$B_{2,N}(s) = \nu_s(\xi_{1,2}\xi_{1,3}), \tag{51}$$

$$B_{3,N}(s) = \nu_s(\xi_{1,2}\xi_{3,4}). \tag{52}$$

At $s = 0$, the replicas are independent and uniform. Expanding ξ_N in distinct Walsh characters and using their orthogonality shows that

$$B_{1,N}(0) = V_N, \qquad B_{2,N}(0) = B_{3,N}(0) = 0. \tag{53}$$

Indeed, the first identity is (27). In each cross term, at least one replica contains a nonconstant Walsh character appearing only once, so its uniform expectation vanishes.

Apply Lemma 2.6 to each of the three observables in (50)–(52). The observables depend on at most four replicas. Every term produced by (48) is N times the expectation of a product of three covariance kernels. By Hölder's inequality, replica symmetry, and Lemma 2.5,

$$\sup_{0\leqslant s\leqslant b^2}|B_{j,N}'(s)| \leqslant CN\sup_{0\leqslant s\leqslant b^2}\nu_s\left(|\xi_N(R_{1,2})|^3\right) \leqslant CN^{1-3p_\star/2}, \tag{54}$$

for $j = 1, 2, 3$. Integrating from 0 to s and using (53), we obtain, uniformly for $0 \leqslant s \leqslant b^2$,

$$B_{1,N}(s) = V_N + O_{b,\xi}\left(N^{1-3p_\star/2}\right) \tag{55}$$

and

$$B_{2,N}(s),\ B_{3,N}(s) = O_{b,\xi}\left(N^{1-3p_\star/2}\right). \tag{56}$$

Substituting (55) and (56) into (49) gives

$$A_N'(s) = NV_N + O_{b,\xi}\left(N^{2-3p_\star/2}\right). \tag{57}$$

Finally, $A_N(0) = 0$ because every interaction character has positive degree and hence zero mean under the uniform measure. Integrating (57) from 0 to s proves (30). ∎

2.4. **Completion of the proof of Theorem 1.1.** We now return to D_N from (24).

First, Gaussian integration by parts gives

$$\frac{d}{ds}\,\mathbb{E}\log Z_N(\sqrt{s}) = \frac{N}{2}\left(\xi_N(1) - A_N(s)\right). \tag{58}$$

On the other hand, from the definition of J_N,

$$\mathbb{E}\big[N J_N(\sqrt{s})\big] = \frac{Ns}{2}\xi_N(1). \tag{59}$$

Differentiating (59) with respect to s, subtracting the resulting identity from (58), and using that both quantities vanish at $s = 0$, we obtain

$$\mathbb{E}\, D_N = -\frac{N}{2}\int_0^{\beta_0^2} A_N(s)\, ds. \tag{60}$$

By Proposition 2.1 and (28),

$$|\,\mathbb{E}\, D_N| \leqslant C_{b,\xi} N^{2-p_\star}. \tag{61}$$

It remains to control the variance. For $p \in \mathcal{P}$ and $I \in \binom{[N]}{p}$, differentiation with respect to the corresponding Gaussian coupling gives

$$\frac{\partial D_N}{\partial g_{p,I}} = \frac{\beta_0\gamma_p}{N^{(p-1)/2}}\langle\sigma_I\rangle_{\beta_0^2} - \frac{\beta_0^2\gamma_p^2}{N^{p-1}} g_{p,I}. \tag{62}$$

Here $\langle\cdot\rangle_{\beta_0^2}$ is the same Gibbs expectation as $\langle\cdot\rangle_{\beta_0}$ in the original inverse-temperature notation. The Gaussian Poincaré inequality therefore gives

$$\mathrm{Var}(D_N) \leqslant \sum_{p\in\mathcal{P}}\sum_{I\in\binom{[N]}{p}} \mathbb{E}\left(\frac{\partial D_N}{\partial g_{p,I}}\right)^2. \tag{63}$$

For each (p, I), Gaussian integration by parts gives

$$\mathbb{E}\left[g_{p,I}\langle\sigma_I\rangle_{\beta_0^2}\right] = \frac{\beta_0\gamma_p}{N^{(p-1)/2}}\,\mathbb{E}\left[1 - \langle\sigma_I\rangle_{\beta_0^2}^2\right]. \tag{64}$$

Expanding the square in (63), using (64), and recalling (29) and (27), we obtain

$$\begin{aligned}\mathrm{Var}(D_N) \leqslant{}& \beta_0^2 N A_N(\beta_0^2) - \beta_0^4 N^2 V_N \\ &+ 2\beta_0^4 \sum_{p\in\mathcal{P}} \frac{\gamma_p^4}{N^{2p-2}} \sum_{I\in\binom{[N]}{p}} \mathbb{E}\langle\sigma_I\rangle_{\beta_0^2}^2.\end{aligned} \tag{65}$$

Proposition 2.1 gives the cancellation

$$\beta_0^2 N A_N(\beta_0^2) - \beta_0^4 N^2 V_N = O_{b,\xi}\big(N^{3-3p_\star/2}\big). \tag{66}$$

For the last term in (65), note that

$$\begin{aligned}
&\sum_{p\in\mathcal{P}} \frac{\gamma_p^4}{N^{2p-2}} \sum_{I\in\binom{[N]}{p}} \mathbb{E}\langle\sigma_I\rangle_{\beta_0^2}^2 \\
&\qquad \leqslant \left(\max_{p\in\mathcal{P}} \frac{\gamma_p^2}{N^{p-1}}\right) \sum_{p\in\mathcal{P}} \frac{\gamma_p^2}{N^{p-1}} \sum_{I\in\binom{[N]}{p}} \mathbb{E}\langle\sigma_I\rangle_{\beta_0^2}^2 \\
&\qquad = \left(\max_{p\in\mathcal{P}} \frac{\gamma_p^2}{N^{p-1}}\right) N A_N(\beta_0^2). \qquad (67)
\end{aligned}$$

By Proposition 2.1 and (28),

$$A_N(\beta_0^2) = O_{b,\xi}(N^{1-p_\star}),$$

while

$$\max_{p\in\mathcal{P}} \frac{\gamma_p^2}{N^{p-1}} = O_\xi(N^{1-p_\star}).$$

Hence the right-hand side of (67) is $O_{b,\xi}(N^{3-2p_\star})$, which is smaller than $N^{3-3p_\star/2}$ because $p_\star > 2$. Combining this with (66) yields

$$\operatorname{Var}(D_N) \leqslant C_{b,\xi} N^{3-3p_\star/2}. \qquad (68)$$

Finally, by (25),

$$\begin{aligned}
&\mathbb{E}\left|N^{p_\star/2}\big(F_N(\beta_0) - J_N(\beta_0)\big)\right|^2 \\
&\qquad = N^{p_\star-2}\left(\operatorname{Var}(D_N) + (\mathbb{E}\, D_N)^2\right) \\
&\qquad \leqslant C_{b,\xi}\left(N^{1-p_\star/2} + N^{2-p_\star}\right) \longrightarrow 0, \qquad (69)
\end{aligned}$$

where we used (61) and (68). The constant depends only on b and the fixed mixture. Since $p_\star \geqslant 3$ and $N \geqslant 1$, one has $N^{2-p_\star} \leqslant N^{1-p_\star/2}$. Taking the supremum over the arbitrary evaluation point $\beta_0 \in [0, b]$ therefore proves (18). At $\beta_0 = 0$ both F_N and J_N vanish identically. The limit (19) follows because $1 - p_\star/2 < 0$. This completes the proof of Theorem 1.1.

3. Proof of the central limit theorem

The quadratic approximation reduces the free-energy fluctuation to an independent sum of centered Gaussian squares. Only the smallest active degree contributes at the stated scale.

Proof of Theorem 1.2. Fix $0 < \beta < \beta_c(\xi)$ and choose b with $\beta \leqslant b < \beta_c(\xi)$. By definition,

$$N^{p_\star/2}\big(J_N(\beta) - \mathbb{E}\, J_N(\beta)\big) = \frac{\beta^2}{2} \sum_{p\in\mathcal{P}} \gamma_p^2 N^{p_\star/2-p} \sum_{I\in\binom{[N]}{p}} (g_{p,I}^2 - 1)\,. \qquad (70)$$

For $p = p_\star$, write $m_N = \binom{N}{p_\star}$. The m_N summands $g_{p_\star,I}^2 - 1$ are independent and identically distributed, with mean zero and variance two. Since $m_N \to \infty$, the classical independent-sum central limit theorem gives

$$\frac{1}{\sqrt{2m_N}} \sum_{I\in\binom{[N]}{p_\star}} (g_{p_\star,I}^2 - 1) \Longrightarrow \mathcal{N}(0,1)\,. \qquad (71)$$

Moreover, $N^{-p_\star/2}\sqrt{2m_N} \to \sqrt{2/p_\star!}$. Hence the contribution of $p_\star$ in (70) converges to the centered Gaussian law with variance $\beta^4\gamma_{p_\star}^4/(2p_\star!)$.

For each $p > p_\star$, the variance of its contribution in (70) equals

$$\frac{\beta^4\gamma_p^4}{2}\, N^{p_\star-2p}\binom{N}{p} = O(N^{p_\star-p}) \longrightarrow 0\,. \tag{72}$$

There are finitely many active degrees, so their total contribution tends to zero in L^2. Theorem 1.1 also gives $N^{p_\star/2}(F_N(\beta) - J_N(\beta)) \to 0$ in L^2. Combining these facts with the exact identity $\mathbb{E}\, J_N(\beta) = \beta^2\xi_N(1)/2$ and Slutsky's theorem proves (20). ∎

References

[1] M. Aizenman, J. L. Lebowitz, and D. Ruelle, *Some rigorous results on the Sherrington–Kirkpatrick spin glass model*, Communications in Mathematical Physics **112** (1987), 3–20. Full text.

[2] D. Banerjee and D. Belius, *Fluctuations of the free energy of the mixed p-spin mean field spin glass model* (2021). arXiv:2108.03109.

[3] X. Bardina, D. Márquez-Carreras, C. Rovira, and S. Tindel, *The p-Spin Interaction Model with External Field*, Potential Analysis **21** (2004), no. 4, 311–362, DOI 10.1023/B:POTA.0000034325.04634.f5.

[4] A. Bovier, I. Kurkova, and M. Löwe, *Fluctuations of the free energy in the REM and the p-spin SK models*, The Annals of Probability **30** (2002), no. 2, 605–651, DOI 10.1214/aop/1023481004.

[5] A. Bovier and A. Schertzer, *Fluctuations of the free energy in p-spin SK models on two scales*, Probability Theory and Related Fields **189** (2024), 771–810, DOI 10.1007/s00440-024-01296-y.

[6] S. Chatterjee, *Disorder chaos and multiple valleys in spin glasses* (2009). arXiv:0907.3381.

[7] W.-K. Chen, P. Dey, and D. Panchenko, *Fluctuations of the free energy in the mixed p-spin models with external field*, Probability Theory and Related Fields **168** (2017), no. 1–2, 41–53, DOI 10.1007/s00440-016-0705-5.

[8] W.-K. Chen, *Phase transition in the spiked random tensor with Rademacher prior*, The Annals of Statistics **47** (2019), no. 5, 2734–2756, DOI 10.1214/18-AOS1763.

[9] F. Comets and J. Neveu, *The Sherrington–Kirkpatrick model of spin glasses and stochastic calculus: The high temperature case*, Communications in Mathematical Physics **166** (1995), no. 3, 549–564, DOI 10.1007/BF02099887.

[10] P. S. Dey and T. Kang, *Fluctuations for the Sherrington–Kirkpatrick spin glass model near the critical temperature* (2026). arXiv:2603.05636.

[11] P. S. Dey and Q. Wu, *Hypergraph Counting and Fluctuations of Mixed p-Spin Glass Models at High Temperature*, Communications in Mathematical Physics **407** (2026), DOI 10.1007/s00220-026-05748-5. Article 230.

[12] P. S. Dey and Q. Wu, *Fluctuation Results for Multi-species Sherrington–Kirkpatrick Model in the Replica Symmetric Regime*, Journal of Statistical Physics **185** (2021), no. 3, DOI 10.1007/s10955-021-02835-w. Article 22.

[13] P. S. Dey and Q. Wu, *Mean Field Spin Glass Models Under Weak External Field*, Communications in Mathematical Physics **402** (2023), no. 2, 1205–1258, DOI 10.1007/s00220-023-04742-5.

[14] F. Guerra and F. L. Toninelli, *Central limit theorem for fluctuations in the high temperature region of the Sherrington–Kirkpatrick spin glass model*, Journal of Mathematical Physics **43** (2002), no. 12, 6224–6237, DOI 10.1063/1.1515109.

[15] H. Knöpfel and M. Löwe, *Fluctuations in a p-spin interaction model*, Annales de l'Institut Henri Poincaré, Probabilités et Statistiques **41** (2005), no. 4, 807–815, DOI 10.1016/j.anihpb.2004.05.006.

[16] D. Panchenko, *The Parisi formula for mixed p-spin models*, The Annals of Probability **42** (2014), no. 3, 946–958, DOI 10.1214/12-AOP800.

[17] M. Talagrand, *Rigorous low-temperature results for the mean field p-spins interaction model*, Probability Theory and Related Fields **117** (2000), no. 3, 303–360, DOI 10.1007/s004400050009.

[18] M. Talagrand, *The Parisi formula*, Annals of Mathematics **163** (2006), no. 1, 221–263, DOI 10.4007/annals.2006.163.221.

[19] S. Tindel, *On the stochastic calculus method for spins systems*, The Annals of Probability **33** (2005), no. 2, 561–581, DOI 10.1214/009117904000000919.